

Bund #1 [West]

NEW · ZONED

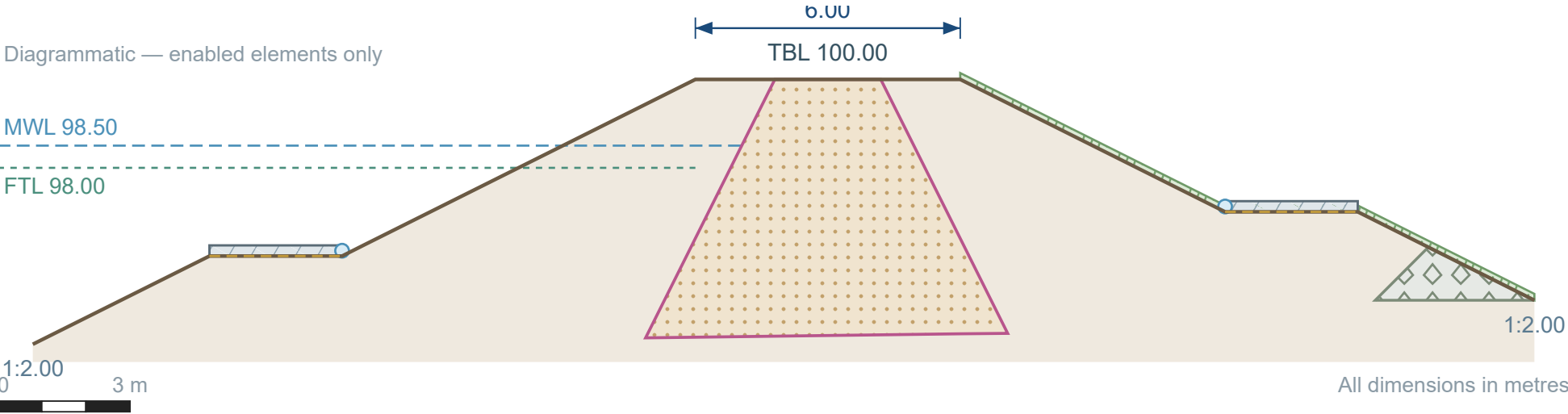
Test reservoir · 3 surveyed section(s) · Bund length 60 m · Datum 0 m · Crest width 6 m

MWL (m)	Freeboard (m)	TBL (m)	FTL (m)	U/S slope	D/S slope
98.5	1.5	100	98	2:1	2:1

Hearding top RL (m)	Hearding top width (m)	U/S hearding slope	D/S hearding slope	Cut-off depth (m)	Cut-off area (m²)	Cut-off bottom width (m)
100	2.4	0.5:1	0.5:1	2	8	3

Casing and impervious hearding are measured separately. The core is contained within the proposed embankment.

Diagrammatic — enabled elements only



Statement of quantities

Average section value = (A1 + A2) / 2. Interval quantity = average section value × length.

Chainage	Length (m)	Average toe RL (m)	Bund height (m)	Bund width / surveyed ground perimeter (m)	Jungle clearance			Bund foundation excavation — depth 0.3 m			Casing zone (earthwork fill)		
					Section	Average	Total	Section	Average	Total	Section	Average	Total
					(m)	(m)	(m²)	(m²)	(m²)	(m³)	(m²)	(m²)	(m³)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
0	—	94.5*	5.8	34.003	34	—	—	10.2	—	—	75.015	—	—
30	30	95*	5.3	32.002	32	33	990	9.6	9.9	297	71.506	73.261	2197.83
60	30	95	5.3	32	32	32	960	9.6	9.6	288	62.34	66.923	2007.69
Total	60						1950			585			4205.52

Statement of quantities — continued

Chainage	Length (m)	Hearting zone (impervious core fill)			D/S turfing			Rock toe			Rock-toe foundation excavation		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m)	(m)	(m ²)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)
0	—	28.328	—	—	8.497	—	—	2.16	—	—	0	—	—
30	30	28.134	28.231	846.93	8.497	8.497	254.91	2.16	2.16	64.8	0	0	0
60	30	24.5	26.317	789.51	8.497	8.497	254.91	2.16	2.16	64.8	0	0	0
Total	60			1636.44			509.82			129.6			0

Statement of quantities — continued

Chainage	Length (m)	Rock-toe graded filter			Chute-drain excavation			Chute-drain lining			Berm filling (U/S, RL 96)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)	(m ³ /m)	(m ³ /m)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(27)	(28)	(29)	(30)	(31)	(32)	(33)	(34)	(35)	(36)	(37)	(38)
0	—	4.449	—	—	0.067	—	—	0.045	—	—	5.096	—	—
30	30	4.449	4.449	133.47	0.067	0.067	2.012	0.045	0.045	1.342	3.956	4.526	135.78
60	30	4.449	4.449	133.47	0.067	0.067	2.012	0.045	0.045	1.342	3	3.478	104.34
Total	60			266.94			4.024			2.684			240.12

Statement of quantities — continued

Chainage	Length (m)	Berm protection (U/S, RL 96)			Berm drain protection (U/S, RL 96)			Berm drain excavation (U/S, RL 96)			Berm filling (D/S, RL 97)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ²)	(m)	(m)	(m ³)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(39)	(40)	(41)	(42)	(43)	(44)	(45)	(46)	(47)	(48)	(49)	(50)
0	—	2.4	—	—	0.12	—	—	0.18	—	—	6.957	—	—
30	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4	6.956	6.957	208.71
60	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4	6	6.478	194.34
Total	60			144			7.2			10.8			403.05

Statement of quantities — continued

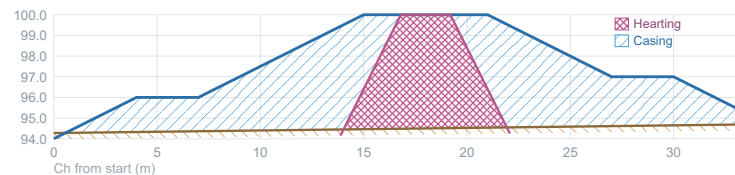
Chainage	Length (m)	Berm protection (D/S, RL 97)			Berm drain protection (D/S, RL 97)			Berm drain excavation (D/S, RL 97)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ²)	(m)	(m)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(51)	(52)	(53)	(54)	(55)	(56)	(57)	(58)	(59)
0	—	2.4	—	—	0.12	—	—	0.18	—	—
30	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4
60	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4
Total	60			144			7.2			10.8

* Detailed existing-ground cross-section; bund width is the surveyed ground perimeter between the designed toes.

Cross-sections

Ch 0 · Average toe RL 94.5 m

GROUND PERIMETER	STRIPPING AREA
34.003 m	10.2 m ²



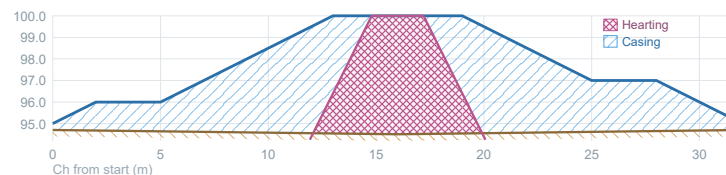
Ch	EL	RL	Calculation	Quantity
0	94.275	94	Start point	
0.564	94.282	94.282	$0.564 \times [(-0.275) + (0)] / 2$	-0.078
4	94.325	96	$3.436 \times [(0) + (1.675)] / 2$	2.878
7	94.363	96	$3 \times [(1.675) + (1.637)] / 2$	4.969
7.5	94.369	96.25	$0.5 \times [(1.637) + (1.881)] / 2$	0.88
15	94.463	100	$7.5 \times [(1.881) + (5.537)] / 2$	27.82
18	94.5	100	$3 \times [(5.537) + (5.5)] / 2$	16.556
21	94.538	100	$3 \times [(5.5) + (5.462)] / 2$	16.444
27	94.613	97	$6 \times [(5.462) + (2.388)] / 2$	23.55
27.5	94.619	97	$0.5 \times [(2.388) + (2.381)] / 2$	1.192
30	94.65	97	$2.5 \times [(2.381) + (2.35)] / 2$	5.914
34	94.7	95	$4 \times [(2.35) + (0.3)] / 2$	5.3
Total formation quantity				105.425 m ²

Impervious heating Casing = formation less heating.

Ch	EL	RL	Calculation	Qty
13.874	94.148	94.148	Start point	
14.025	94.15	94.45	$0.151 \times [(0) + (0)] / 2$	0
15	94.163	96.4	$0.975 \times [(0) + (1.937)] / 2$	0.945
16.8	94.185	100	$1.8 \times [(1.937) + (5.515)] / 2$	6.707
18	94.2	100	$1.2 \times [(5.515) + (5.5)] / 2$	6.609
19.2	94.215	100	$1.2 \times [(5.5) + (5.485)] / 2$	6.591
21	94.238	96.401	$1.8 \times [(5.485) + (1.863)] / 2$	6.613
21.926	94.249	94.549	$0.926 \times [(1.863) + (0)] / 2$	0.863
22.075	94.251	94.251	$0.149 \times [(0) + (0)] / 2$	0

Ch 30 · Average toe RL 95 m

GROUND PERIMETER	STRIPPING AREA
32.002 m	9.6 m ²



Ch	EL	RL	Calculation	Quantity
0	94.7	95	Start point	
2	94.675	96	$2 \times [(0.3) + (1.325)] / 2$	1.625
5	94.638	96	$3 \times [(1.325) + (1.362)] / 2$	4.031
6.5	94.619	96.75	$1.5 \times [(1.362) + (2.131)] / 2$	2.62
13	94.538	100	$6.5 \times [(2.131) + (5.463)] / 2$	24.68
16	94.5	100	$3 \times [(5.463) + (5.5)] / 2$	16.444
19	94.538	100	$3 \times [(5.5) + (5.463)] / 2$	16.444
25	94.613	97	$6 \times [(5.463) + (2.388)] / 2$	23.55
25.5	94.619	97	$0.5 \times [(2.388) + (2.381)] / 2$	1.192
28	94.65	97	$2.5 \times [(2.381) + (2.35)] / 2$	5.914
32	94.7	95	$4 \times [(2.35) + (0.3)] / 2$	5.3
Total formation quantity				101.8 m ²

Impervious heating Casing = formation less heating.

Ch	EL	RL	Calculation	Qty
11.925	94.251	94.251	Start point	
12.074	94.249	94.549	$0.149 \times [(0) + (0)] / 2$	0
13	94.238	96.401	$0.926 \times [(0) + (1.863)] / 2$	0.863
14.8	94.215	100	$1.8 \times [(1.863) + (5.485)] / 2$	6.613
16	94.2	100	$1.2 \times [(5.485) + (5.5)] / 2$	6.591
17.2	94.215	100	$1.2 \times [(5.5) + (5.485)] / 2$	6.591
19	94.238	96.401	$1.8 \times [(5.485) + (1.863)] / 2$	6.613
19.926	94.249	94.549	$0.926 \times [(1.863) + (0)] / 2$	0.863
20.075	94.251	94.251	$0.149 \times [(0) + (0)] / 2$	0

Earthwork excavation classification

Percentages are the selected dashboard classification applied to each measured source.

Foundation / stripping · 585 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$585 \times 100 / 100$	585

Chute drain · 4.024 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$4.024 \times 100 / 100$	4.024

Berm catch-water drains · 21.6 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$21.6 \times 100 / 100$	21.6

Total quantities of Excavation.

Each excavation DATA code is shown once, followed by the works contributing to its total.

IRR-CAW-1-1.

Foundation stripping	Chute drains	Berm catch-water drains	Total
585	+ 4.024	+ 21.6	= <u>610.624</u> m ³

Total quantities

Each work code is shown once, followed by the components that add to its total. Excavation quantities are in Total quantities of Excavation above and are not repeated here. Distinct operations can share a measured volume; their quantities must not be added to obtain geometric fill.

IRR-PMW-1-2.

Jungle clearance	Total
1950	= <u>1950</u> m ²

IRR-DAW-5-3.

Casing	u/s Berm (RL 96)	d/s Berm (RL 97)	Less: Rock toe	Total
3691.95	+ 240.12	+ 403.05	-129.6	= <u>4205.52</u> m ³

IRR-PMW-3-8.

Zoned embankment - hearting formation	Total
1636.44	= <u>1636.44</u> m ³

IRR-DAW-6-15.

Bund	Total
509.82	= <u>509.82</u> SQM

IRR-CAW-7-15.

D/S chute drains — channel protection	Total
2.684	= <u>2.684</u> CUM

IRR-CAW-8-15.

u/s Berm (RL 96)	Total
288	= <u>288</u> SQM

IRR-CAW-7-12.

u/s Berm (RL 96)	Total
14.4	= <u>14.4</u> CUM

IRR-CAW-5-6.

Rock toe (d/s)	Total
129.6	= <u>129.6</u> CUM

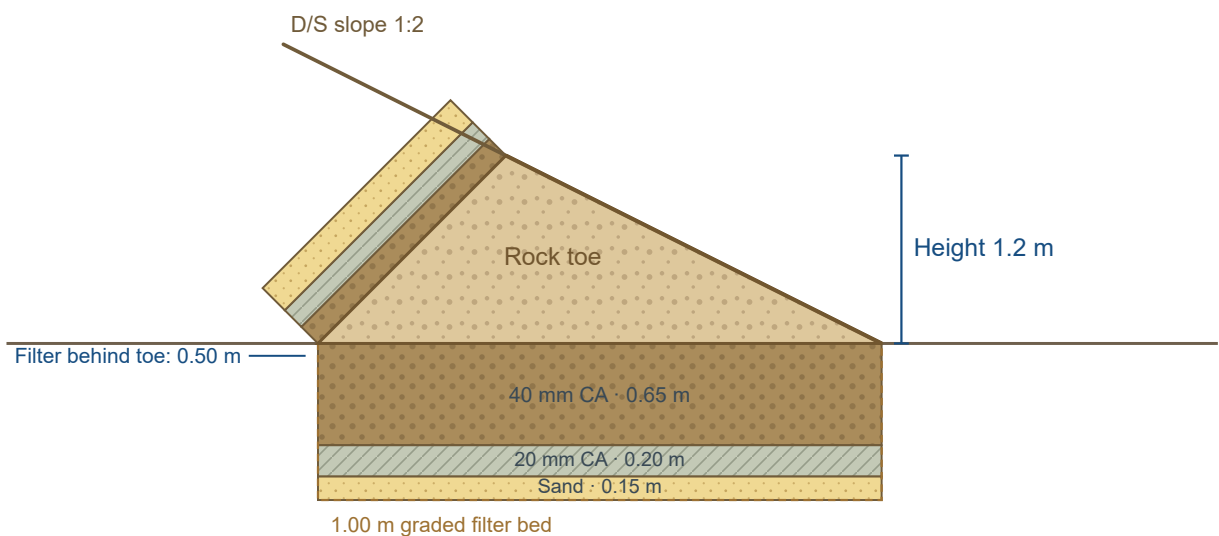
IRR-CAW-5-11.

Graded filter below and behind rock toe (d/s)	Total
266.94	= <u>266.94</u> CUM

Component details

Rock toe and graded filter

The shared general excavation beneath the rock-toe foundation is assigned to the rock-toe trench and deducted from the general cut. It is counted once.



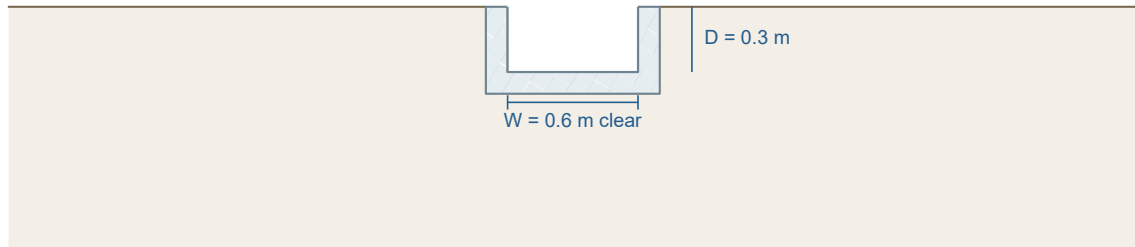
$$\text{Area} = H \times [\text{Crest} + H \times (\text{inner} + \text{outer}) / 2] = 1.2 \times [0 + 1.2 \times (1 + 2) / 2] = 2.16 \text{ m}^2$$

$$\text{Base } 3.6 \text{ m} \cdot \text{inner } 1:1 \cdot \text{outer} = \text{D/S slope}$$

Chute drains

Excavation area: $A = W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$. Lined perimeter: $P = W + 2D = 1.2 \text{ m}$. Per chute, excavation = $A \times \text{developed length}$; protection = $P \times \text{developed length} \times \text{lining thickness}$.

CC protection: $t = 0.1 \text{ m}$



Excavation area = $W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$

Lined perimeter $P = W + 2D = 0.6 + 2(0.3) = 1.2 \text{ m}$

Cross-section of downstream chute drain

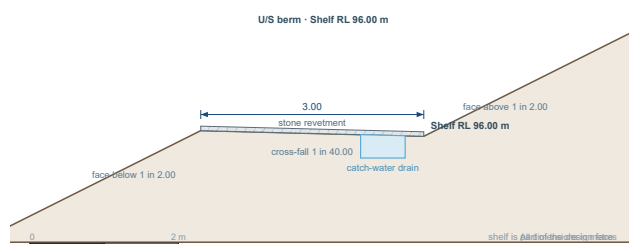
Berm shelves

U/S shelves sit on the left (water face, mirrored); D/S shelves sit on the right. Each figure is labelled with its shelf RL.

U/S berms

U/S berm · Shelf RL 96 m

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.

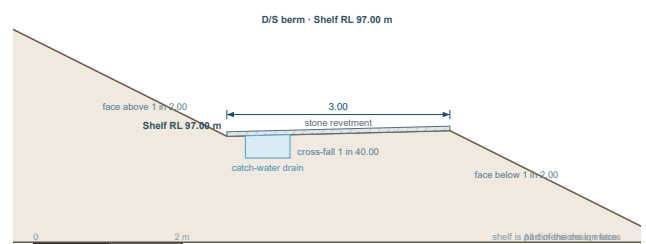


Shelf surfacing: 144 m^2 Drain protection: 7.2 m^3 Drain excavation: 10.8 m^3

D/S berms

D/S berm · Shelf RL 97 m

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.



Shelf surfacing: 144 m^2 Drain protection: 7.2 m^3 Drain excavation: 10.8 m^3